

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250274842

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Dreiling; Ryan P. et al.

SMART DEVICE APPLICATION PRIORITIZATION FOR SATELLITE NETWORKS

Abstract

The wireless device connects to a non-terrestrial communication network from a terrestrial communication network such that a wireless communication service is provided by the non-terrestrial communication network. The wireless device receives an indication of resource constraints of the non-terrestrial communication network from the non-terrestrial communication network. The wireless device determines a list of applications on the wireless device that comply with the resource constraints of the non-terrestrial communication network. The wireless device restricts access to the applications that exceed the resource constraints of the non-terrestrial communication network. The wireless device prioritizes the applications on the wireless device that comply with the resource constraints of the non-terrestrial communication network, where the prioritization is unique to each wireless device, location, or time. The wireless device updates the list of applications and prioritization of each application based on received changes to the resource constraints of the non-terrestrial communication network.

Inventors: Dreiling; Ryan P. (Shawnee, KS), Ives; Derrick L. (North Bend, WA), Lucht; Thomas P. (Seattle, WA), Shah; Mehul (Bellevue, WA), Wang; Helen (Bellevue, WA)

Applicant: T-Mobile USA, Inc. (Bellevue, WA)

Family ID: 1000007829651

Appl. No.: 18/590762

Filed: February 28, 2024

Publication Classification

Int. Cl.: H04W48/02 (20090101); H04W48/16 (20090101); H04W72/563 (20230101); H04W84/06 (20090101)

U.S. Cl.:

CPC **H04W48/02** (20130101); **H04W48/16** (20130101); **H04W72/563** (20230101);
H04W84/06 (20130101)

Background/Summary

BACKGROUND

[0001] Current wireless communications systems (e.g., fifth-generation mobile network (5G)) utilize base stations to communicate with a user equipment. Base stations can be located at the surface of the Earth and support telecommunications coverage in a surrounding area. When in a coverage region of the base station, the user equipment can connect with the base station to communicate data through the network. A new approach is being pursued to use non-terrestrial network, with base stations in orbiting satellites. The user equipment can connect to the satellite when within a coverage region of the satellite. In general, a satellite can provide a larger coverage region and can more easily provide coverage to remote locations. Accordingly, network providers are utilizing non-terrestrial networks to increase coverage and provide improved networks.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Detailed descriptions of implementations of the present invention will be described and explained through the use of the accompanying drawings.

[0003] FIG. 1 illustrates a wireless communications system that can implement aspects of the present technology.

[0004] FIG. 2 illustrates a method for configuring device settings when connected with a satellite network in accordance with aspects of the present technology.

[0005] FIG. 3 illustrates device configuration data in accordance with aspects of the present technology.

[0006] FIG. 4 is a block diagram that illustrates a wireless device prioritizing applications when connected to a satellite network.

[0007] FIG. 5 is a flowchart that illustrates a process for restricting and prioritizing applications when connected to a non-terrestrial communication network.

[0008] FIG. 6 illustrates components of a computing device that can implement aspects of the present technology.

[0009] The technologies described herein will become more apparent to those skilled in the art from studying the Detailed Description in conjunction with the drawings. Embodiments or implementations describing aspects of the invention are illustrated by way of example, and the same references can indicate similar elements. While the drawings depict various implementations for the purpose of illustration, those skilled in the art will recognize that alternative implementations can be employed without departing from the principles of the present technologies. Accordingly, while specific implementations are shown in the drawings, the technology is amenable to various modifications.

DETAILED DESCRIPTION

[0010] Given that satellites are not bound to the surface of the Earth, satellites can provide a larger coverage region than base stations and more easily provide coverage in remote locations. As a consequence of this increased coverage region, a greater number of users may compete for communication resources provided by the satellite networks, thereby increasing congestion. This

congestion can be worsened by the limited wireless resources that are provided by these networks. For example, satellite networks can be difficult or costly to implement, resulting in these networks having decreased bandwidths or greater latency. Thus, satellite networks can be resource-constrained due to increased competition for limited communication resources.

[0011] Given that satellite networks and terrestrial networks have different capabilities, different device settings can enable wireless devices to function more efficiently on different networks. For example, satellite networks can be more resource-constrained than terrestrial networks implemented through surface-bound base stations and thus, mobile network providers can choose to disable some wireless services on satellite networks. A satellite network can have less capacity, be more susceptible to congestion, and have limited functionality compared to terrestrial networks.

[0012] For example, mobile network providers may choose to disable the use of certain applications on a wireless device when on satellite networks. Applications can include software applications developed specifically for use on small, wireless computing devices, such as smartphones and tablets. The use of certain applications by the wireless device on the satellite network may not be suitable due to the amount of resources the applications consume. Applications that consume large amounts of resources on the satellite network can limit the available capacity on the satellite network, which can limit access to important network services, such as communication services, for other users.

[0013] The mobile network providers can change the wireless device settings to prevent the wireless device from accessing applications that require high amounts of resources. When connected to the satellite network, the wireless device can receive an indication of the resource constraints of the satellite network from the satellite network. The wireless device can determine a list of applications found on the wireless device that comply with the received resource constraints. Access can be restricted to applications that do not comply with the resource constraints of the network. Applications that comply with the resource constraints can be prioritized on the wireless device. For example, prioritization can be based on recency of use, frequency of use, or the amount of resources the application requires. The prioritization can be unique to each wireless device, location, or time. Updated resource constraints of the satellite network can be received by the wireless device, allowing the wireless device to update the list of applications that comply with the resource constraints and the prioritization of the applications that comply with the resource constraints.

[0014] The device settings can be determined using information that is broadcast by available networks that have a coverage area that encompasses the location of a wireless device. For example, networks can broadcast an associated public land mobile network (PLMN) code, including a Mobile Country Code (MCC) and Mobile Network Code (MNC), within system information blocks. The wireless device can receive the system information blocks and determine if the wireless device complies with certain device settings when connected to a network associated with the PLMN code. In some cases, the wireless device is to comply with certain device settings only when the network is of a particular type. For example, the wireless device can determine the appropriate device settings, such as restricting the wireless device to text-based communications only when the network is determined to be a satellite network.

[0015] Alternatively, or additionally, the wireless device is to comply with certain device settings only when the network is associated with a particular network provider. For example, a network provider can provide device settings for different networks associated with the particular network provider. As a specific example, a satellite network partnered with a terrestrial network provider can be allocated a PLMN code belonging to the terrestrial network provider rather than provisioning an entirely new PLMN code to the satellite network. As discussed above, satellite networks can have different capabilities than terrestrial networks. Thus, the terrestrial network provider can benefit from specifying different device settings associated with different networks having PLMN codes belonging to the terrestrial network provider. Given that the device settings

are associated with networks having PLMN codes belonging to the terrestrial network provider, in some cases, the device settings can be determined in response to disconnecting from the terrestrial network of a provider and connecting to the non-terrestrial network of the same provider. [0016] The description and associated drawings are illustrative examples and are not to be construed as limiting. This disclosure provides certain details for a thorough understanding and enabling description of these examples. One skilled in the relevant technology will understand, however, that the invention can be practiced without many of these details. Likewise, one skilled in the relevant technology will understand that the invention can include well-known structures or features that are not shown or described in detail to avoid unnecessarily obscuring the descriptions of examples.

Wireless Communications System

[0017] FIG. 1 is a block diagram that illustrates a wireless telecommunication network **100** (“network **100**”) in which aspects of the disclosed technology are incorporated. The network **100** includes base stations **102-1** through **102-4** (also referred to individually as “base station **102**” or collectively as “base stations **102**”). A base station is a type of network access node (NAN) that can also be referred to as a cell site, a base transceiver station, or a radio base station. The network **100** can include any combination of NANs including an access point, radio transceiver, gNodeB (gNB), NodeB, eNodeB (eNB), Home NodeB or Home eNodeB, or the like. In addition to being a wireless wide area network (WWAN) base station, a NAN can be a wireless local area network (WLAN) access point, such as an Institute of Electrical and Electronics Engineers (IEEE) 802.11 access point.

[0018] The NANs of a network **100** formed by the network **100** also include wireless devices **104-1** through **104-7** (referred to individually as “wireless device **104**” or collectively as “wireless devices **104**”) and a core network **106**. The wireless devices **104** can correspond to or include network **100** entities capable of communication using various connectivity standards. For example, a 5G communication channel can use millimeter wave (mmW) access frequencies of 28 GHz or more. In some implementations, the wireless device **104** can operatively couple to a base station **102** over a long-term evolution/long-term evolution-advanced (LTE/LTE-A) communication channel, which is referred to as a 4G communication channel.

[0019] The core network **106** provides, manages, and controls security services, user authentication, access authorization, tracking, internet protocol (IP) connectivity, and other access, routing, or mobility functions. The base stations **102** interface with the core network **106** through a first set of backhaul links (e.g., S1 interfaces) and can perform radio configuration and scheduling for communication with the wireless devices **104** or can operate under the control of a base station controller (not shown). In some examples, the base stations **102** can communicate with each other, either directly or indirectly (e.g., through the core network **106**), over a second set of backhaul links **110-1** through **110-3** (e.g., X1 interfaces), which can be wired or wireless communication links.

[0020] The base stations **102** can wirelessly communicate with the wireless devices **104** via one or more base station antennas. The cell sites can provide communication coverage for geographic coverage areas **112-1** through **112-4** (also referred to individually as “coverage area **112**” or collectively as “coverage areas **112**”). The coverage area **112** for a base station **102** can be divided into sectors making up only a portion of the coverage area (not shown). The network **100** can include base stations of different types (e.g., macro and/or small cell base stations). In some implementations, there can be overlapping coverage areas **112** for different service environments (e.g., Internet of Things (IoT), mobile broadband (MBB), vehicle-to-everything (V2X), machine-to-machine (M2M), machine-to-everything (M2X), ultra-reliable low-latency communication (URLLC), machine-type communication (MTC), etc.).

[0021] The network **100** can include a 5G network and/or an LTE/LTE-A or other network. In an LTE/LTE-A network, the term “eNBs” is used to describe the base stations **102**, and in 5G new radio (NR) networks, the term “gNBs” is used to describe the base stations **102** that can include

mmW communications. The network **100** can thus form a heterogeneous network **100** in which different types of base stations provide coverage for various geographic regions. For example, each base station **102** can provide communication coverage for a macro cell, a small cell, and/or other types of cells. As used herein, the term “cell” can relate to a base station, a carrier or component carrier associated with the base station, or a coverage area (e.g., sector) of a carrier or base station, depending on context.

[0022] A macro cell generally covers a relatively large geographic area (e.g., several kilometers in radius) and can allow access by wireless devices that have service subscriptions with a wireless network **100** service provider. As indicated earlier, a small cell is a lower-powered base station, as compared to a macro cell, and can operate in the same or different (e.g., licensed, unlicensed) frequency bands as macro cells. Examples of small cells include pico cells, femto cells, and micro cells. In general, a pico cell can cover a relatively smaller geographic area and can allow unrestricted access by wireless devices that have service subscriptions with the network **100** provider. A femto cell covers a relatively smaller geographic area (e.g., a home) and can provide restricted access by wireless devices having an association with the femto unit (e.g., wireless devices in a closed subscriber group (CSG), wireless devices for users in the home). A base station can support one or multiple (e.g., two, three, four, and the like) cells (e.g., component carriers). All fixed transceivers noted herein that can provide access to the network **100** are NANs, including small cells.

[0023] The communication networks that accommodate various disclosed examples can be packet-based networks that operate according to a layered protocol stack. In the user plane, communications at the bearer or Packet Data Convergence Protocol (PDCP) layer can be IP-based. A Radio Link Control (RLC) layer then performs packet segmentation and reassembly to communicate over logical channels. A Medium Access Control (MAC) layer can perform priority handling and multiplexing of logical channels into transport channels. The MAC layer can also use Hybrid ARQ (HARQ) to provide retransmission at the MAC layer, to improve link efficiency. In the control plane, the Radio Resource Control (RRC) protocol layer provides establishment, configuration, and maintenance of an RRC connection between a wireless device **104** and the base stations **102** or core network **106** supporting radio bearers for the user plane data. At the Physical (PHY) layer, the transport channels are mapped to physical channels.

[0024] Wireless devices can be integrated with or embedded in other devices. As illustrated, the wireless devices **104** are distributed throughout the network **100**, where each wireless device **104** can be stationary or mobile. For example, wireless devices can include handheld mobile devices **104-1** and **104-2** (e.g., smartphones, portable hotspots, tablets, etc.); laptops **104-3**; wearables **104-4**; drones **104-5**; vehicles with wireless connectivity **104-6**; head-mounted displays with wireless augmented reality/virtual reality (AR/VR) connectivity **104-7**; portable gaming consoles; wireless routers, gateways, modems, and other fixed-wireless access devices; wirelessly connected sensors that provide data to a remote server over a network; IoT devices such as wirelessly connected smart home appliances; etc.

[0025] A wireless device (e.g., wireless devices **104**) can be referred to as a user equipment (UE), a customer premises equipment (CPE), a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a handheld mobile device, a remote device, a mobile subscriber station, a terminal equipment, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a mobile client, a client, or the like.

[0026] A wireless device can communicate with various types of base stations and network **100** equipment at the edge of a network **100** including macro eNBs/gNBs, small cell eNBs/gNBs, relay base stations, and the like. A wireless device can also communicate with other wireless devices either within or outside the same coverage area of a base station via device-to-device (D2D) communications.

[0027] The communication links **114-1** through **114-9** (also referred to individually as

“communication link **114**” or collectively as “communication links **114**”) shown in network **100** include uplink (UL) transmissions from a wireless device **104** to a base station **102** and/or downlink (DL) transmissions from a base station **102** to a wireless device **104**. The downlink transmissions can also be called forward link transmissions while the uplink transmissions can also be called reverse link transmissions. Each communication link **114** includes one or more carriers, where each carrier can be a signal composed of multiple sub-carriers (e.g., waveform signals of different frequencies) modulated according to the various radio technologies. Each modulated signal can be sent on a different sub-carrier and carry control information (e.g., reference signals, control channels), overhead information, user data, etc. The communication links **114** can transmit bidirectional communications using frequency division duplex (FDD) (e.g., using paired spectrum resources) or time division duplex (TDD) operation (e.g., using unpaired spectrum resources). In some implementations, the communication links **114** include LTE and/or mmW communication links.

[0028] In some implementations of the network **100**, the base stations **102** and/or the wireless devices **104** include multiple antennas for employing antenna diversity schemes to improve communication quality and reliability between base stations **102** and wireless devices **104**.

Additionally or alternatively, the base stations **102** and/or the wireless devices **104** can employ multiple-input, multiple-output (MIMO) techniques that can take advantage of multi-path environments to transmit multiple spatial layers carrying the same or different coded data.

[0029] In some examples, the network **100** implements 6G technologies including increased densification or diversification of network nodes. The network **100** can enable terrestrial and non-terrestrial transmissions. In this context, a non-terrestrial network (NTN) is enabled by one or more satellites, such as satellites **116-1** and **116-2**, to deliver services anywhere and anytime and provide coverage in areas that are unreachable by any conventional terrestrial network (TN). A 6G implementation of the network **100** can support terahertz (THz) communications. This can support wireless applications that demand ultrahigh quality of service (QOS) requirements and multi-terabits-per-second data transmission in the era of 6G and beyond, such as terabit-per-second backhaul systems, ultra-high-definition content streaming among mobile devices, AR/VR, and wireless high-bandwidth secure communications. In another example of 6G, the network **100** can implement a converged Radio Access Network (RAN) and core architecture to achieve Control and User Plane Separation (CUPS) and achieve extremely low user plane latency. In yet another example of 6G, the network **100** can implement a converged Wi-Fi and core architecture to increase and improve indoor coverage.

Configuring a Device for Connection to a Satellite Network

[0030] FIG. **2** illustrates a method **200** for configuring device settings when connecting with a satellite network in accordance with aspects of the present technology. Although illustrated in a particular configuration, one or more operations of the method **200** may be omitted, repeated, or reorganized. Additionally, the method **200** may include other operations not illustrated in FIG. **2**—for example, operations detailed in one or more other methods described herein.

[0031] As illustrated, a wireless device **202** (e.g., an example of the wireless device **104** of FIG. **1**) can be capable of communicating with different network types. For example, the wireless device **202** can connect with terrestrial networks (e.g., a 3G, LTE, 4G, 5G, or other terrestrial network) or non-terrestrial networks. Terrestrial networks can be implemented through any number of base stations located at the surface of the Earth. The terrestrial network can include a home terrestrial network, one or more partnered terrestrial networks, and one or more non-partnered terrestrial networks. The home terrestrial network can have an MCC and MNC that are the same as the MCC and MNC indicated within an international mobile subscriber identity (IMSI) of the wireless device **202**. The partnered terrestrial networks can have a partnership agreement with the home mobile network to provide roaming services to the wireless device **202** within a coverage area of the partnered terrestrial networks. The non-partnered terrestrial networks may not have a partnership

agreement in place with the home terrestrial network but may still provide some limited services to the wireless device **202**, such as emergency services, within a coverage region of the non-partnered terrestrial networks.

[0032] In contrast, satellite networks (or non-terrestrial networks) can be implemented through at least one satellite orbiting the Earth. As illustrated in FIG. 2, the wireless device **202** communicates with at least one satellite network **204**. In aspects, the satellite network **204** includes a satellite network provided by a same mobile network provider as the home terrestrial network or having a partnership agreement with the home terrestrial network. Thus, the satellite network **204** can provide wireless communication services to the wireless device **202** within a coverage region of the satellite network **204**. In aspects, non-partnered satellite networks can provide only limited services to the wireless device **202**. As a result, non-partnered satellite networks can be treated similarly to non-partnered terrestrial networks in that the wireless device **202** generally will not attempt to connect with non-partnered networks when requesting non-emergency services.

[0033] The wireless device **202** includes communication logic **206** capable of controlling the transmission of signaling to a communications network (e.g., the satellite network **204** or a terrestrial network) and the reception of signaling from the communication network using a wireless transceiver. The communication logic **206** can be implemented in hardware, software, or firmware. In aspects, the communication logic **206** can process the signals received at the wireless transceiver in accordance with a communication technology. For example, the communication logic **206** can analyze system information received from available networks to determine if a network is available to provide a wireless communication service to the wireless device **202**. In response to determining that a network is available to provide the service, the communication logic **206** can request to connect to the available network such that the available network can provide a wireless service to the wireless device.

[0034] The communication logic **206** can provide network information (e.g., located within system information blocks broadcast by proximate networks) to a device settings application **208** responsible for configuring the device to have particular device settings with which the wireless device **202** is to comply when connected with the available network. The device settings application **208** can utilize information about the available network to determine particular device settings for the wireless device **202** when the wireless device **202** is connected to the available network. The device settings application **208** can retrieve the device settings from device configuration data. The device configuration data can store sets of device settings with which the wireless device **202** is to comply when connected to respective networks associated with respective entries of network information in the device configuration data. The device settings can provide the wireless device **202** permission to request one or more wireless services from a network, prevent the wireless device **202** from requesting one or more wireless services from the network, or adjust a visual, auditory, or haptic characteristic of the wireless device **202**. In some cases, the device configuration data can be associated with a particular mobile network provider to differentiate between sets of device settings with which the wireless device is to comply when connected to the different networks provided by the mobile network provider. In some cases, the device configuration data can include device settings associated with different types of networks (e.g., non-terrestrial networks, terrestrial networks, generations of these networks, and so on) such that the device settings application **208** can use the device configuration data to determine the device settings with which the wireless device **202** is to comply when connected to a particular type of network. Further details of the device configuration data can be described with respect to FIG. 3.

[0035] At **210**, the wireless device **202** can receive broadcast information related to and broadcasted by a proximate network (e.g., a network whose coverage area includes the location of the wireless device). As illustrated, the satellite network **204** transmits network information that is received by the wireless device **202**. For example, the satellite network **204** can transmit a PLMN code allocated to the satellite network **204** within system information blocks broadcast by the

satellite network **204**. The network information related to the satellite network **204** can be broadcast repeatedly at predetermined intervals. The wireless device **202** can receive the network information from the satellite network **204** when within a coverage region of the satellite network **204**. For example, the communication logic **206** can tune a receiver of the wireless device **202** to receive the network information from the satellite network **204**. If the wireless device **202** is outside of the coverage region of the satellite network **204**, the wireless device **202** may be unable to receive the network information.

[0036] The communication logic **206** can analyze the network information to select a network to which the wireless device **202** is to connect. The network information can be analyzed at least once every predetermined period of time to enable network selection/re-selection. As a specific example, network information can be analyzed at least once every six minutes to enable network selection/re-selection once every six minutes.

[0037] The communication logic **206** can compare the network information to one or more lists of approved networks to which the wireless device **202** is approved to connect. In aspects, the approved networks can include a home terrestrial network of the wireless device **202** and at least one partnered network partnered with the home network to allow the wireless device **202** to roam on the partnered network. In some cases, the partnered network can include a partnered satellite network, such as the satellite network **204**. The information associated with one or more of the approved networks can be stored in a list of approved mobile networks. Thus, the network information can be compared to the PLMN code of the home network and one or more PLMN codes of partnered networks to determine if the network information is associated with one or more approved networks. If so, the wireless device **202** can determine that an approved network is available to the wireless device **202**.

[0038] At **212**, the wireless device **202** can initiate a wireless access request to an approved network that is available to provide the wireless communication service. The wireless access request can be initiated in response to determine that an approved network is available to provide the wireless communication service. The wireless access request can be initiated by performing at least a portion of a network selection/re-selection procedure. For example, network information received from the satellite network **204** can be used to determine the satellite network **204** as an approved network. As a result, the wireless device **202** can initiate the wireless access request to the satellite network **204** to request the satellite network **204** to provide the wireless communication service to the wireless device **202**. In response, the wireless device **202** can connect to the satellite network **204** such that the satellite network **204** provides the wireless service (e.g., message, call, data, or other services) to the wireless device **202**.

[0039] At **214**, the communication logic **206** can provide an indication of the network information to the device settings application **208**. In some cases, the communication logic **206** only provides the device settings application **208** the indication of the network information once it is determined that the wireless device **202** will attempt to connect to the network. For example, the network information can be indicated to the device settings application **208** once the satellite network **204** is determined as an approved network capable of providing the wireless service. In some cases, the indication of the network information can be provided to the device settings application **208** after transmission of the network connection request.

[0040] At **216**, the device settings application **208** determines the type of network with which the wireless device **202** is to connect. The determination can be based on the indication of the network information. In some cases, the network information can include the PLMN code of the network and determine the network type based on the PLMN code. For example, the wireless device **202** can compare the PLMN code to data that associates different PLMN codes with different network types. In doing so, the wireless device **202** can determine that the satellite network **204** is a non-terrestrial network. In response to determining that the satellite network **204** is a non-terrestrial network, the device settings application **208** can determine appropriate device settings for the

wireless device **202** when connected with the satellite network **204** at **220**.

[0041] Alternatively, or additionally, at **218**, the device settings application **208** can determine a mobile network provider associated with the network information. As discussed above, the satellite network **204** can be partnered with a home network of the wireless device **202**. In some cases, the PLMN code of the satellite network **204** can be allocated to the satellite network **204** by a network provider of the home network of the wireless device **202**. For example, the network provider of the home network of the wireless device **202** can have multiple associated PLMN codes, and the network provider can allocate one or more of the PLMN codes to the satellite network **204**. The device settings application **208** can compare the PLMN code of the satellite network **204** to data associating PLMN codes with different mobile network providers. In doing so, the device settings application **208** can determine that the network information is associated with the mobile network provider of the home network. After determining that the network information is associated with the mobile network provider, the device settings application **208** can compare the network information to device configuration data that associates entries of network information of networks associated with the mobile network provider to different network types or sets of device settings. In this way, the device settings application **208** can determine appropriate settings for the wireless device **202** when connected with the satellite network **204** at **220**.

[0042] Although the determinations at **216** and **218** are illustrated and described as being performed at the device settings application **208**, in other cases, the determination at **216** and **218** can be performed at the communication logic **206** or any other component and an indication of the determination can be provided to the device settings application **208**.

[0043] At **220**, the device settings application **208** determines one or more device settings associated with the network information. The device settings application **208** can determine the device settings by comparing the network information to a device configuration data. The device configuration data can be maintained on the SIM or on any other storage within or coupled with the wireless device **202**. Alternatively, the device settings application **208** can receive the device configuration data from the satellite network **204** or any other wireless network using a receiver of the wireless device **202** and the communication logic **206**. In aspects, the device configuration data can include one or more entries of network information, respective network types of respective ones of the entries of network information, or respective device settings associated with the respective ones of the entries of network information. Thus, by comparing the network information to the device configuration data, device settings with which the wireless device **202** is to comply when connected with the satellite network **204** can be determined.

[0044] In some embodiments, the device settings are determined after the satellite network **204** is determined to have a particular type that may require particular device settings. For example, the device settings application **208** can determine the device settings with which the wireless device **202** is to comply when connected with the satellite network **204** after determining that the satellite network **204** is a non-terrestrial network. In some cases, the device configuration data can associate a single set of device settings with a network type. In this way, the device settings can be determined by searching for the set of settings corresponding to the network type. In other cases, the device settings can be assigned on a per-network basis such that two different non-terrestrial networks can have different sets of device settings. Thus, the specific network information can be used to determine the appropriate set of device settings with which to comply when connected with the satellite network **204**.

[0045] In some embodiments, the device settings are determined after the network information is determined to be associated with a particular provider. For example, the satellite network **204** can be allocated a PLMN code (e.g., the satellite network **204** broadcasts the PLMN code) that belongs to a first mobile network provider (e.g., is associated with the first mobile network provider in a PLMN code list) with which a second provider of the satellite network **204** is partnered. In aspects, the second provider of the satellite network **204** can have a partnership agreement with the first

mobile network provider such that customers of the first mobile network provider can communicate on the satellite network **204**. Given that the network information is associated with the first mobile network provider and allocated, by the first mobile network provider, to the satellite network **204**, when the wireless device **202** analyzes the PLMN code of the satellite network **204**, it can determine that the PLMN code belongs to the first mobile network provider. The first mobile network provider can allocate PLMN codes to different networks that operate better with different device settings. To accommodate the various sets of device settings associated with the different networks, the first mobile network provider can provide the wireless device **202** with device configuration data that includes various network information associated with respective device settings, and the wireless device **202** can reference the device configuration data once the network information is determined to be associated with the first mobile network provider.

[0046] At **222**, the device settings application **208** can configure the wireless device **202** in accordance with the device settings determined at **220**. Configuring the wireless device **202** in accordance with the device settings can be performed before or after connecting with the satellite network **204**. In aspects, the wireless device **202** can be configured such that the wireless device **202** is unable to request one or more wireless services from the satellite network **204**. For example, the wireless device **202** can be disabled from requesting voice call or data services from the satellite network **204** due to the reduced resources available on the satellite network. In doing so, not only can network resources be saved by not providing the wireless services, but also network resources can be saved by not having to communicate the connection requests for these services, which can be particularly beneficial in resource-constrained satellite networks. In some embodiments, the device settings can relate to a visual, auditory, or haptic characteristic of the wireless device **202**. In this way, configuring the wireless device **202** to comply with the device settings can include adjusting a visual, auditory, or haptic characteristic of the wireless device **202**. As specific examples, the wireless device **202** can display an icon indicative of a connection to the satellite network **204** on a display of the wireless device **202**, output a sound indicative of a connection to the satellite network **204** using a speaker of the wireless device **202**, or actuate a haptic element of the wireless device **202** to produce haptic feedback at the wireless device **202**.

[0047] Although specific examples are provided, the method **200** can be performed to configure a wireless device **202** in accordance with different device settings. Moreover, the device settings application **208** can configure the wireless device **202** in response to different networks or different types of networks. Thus, from the foregoing, it is appreciated that the particular examples described with respect to FIG. **2** are but some of many possible examples.

[0048] FIG. **3** illustrates device configuration data **300** in accordance with aspects of the present technology. As discussed above, the device configuration data **300** can include one or more entries of network information and respective ones of one or more device settings associated with respective entries of network information. The device configuration data **300** can be stored in any number of data structures (e.g., a table, linked list, multi-dimensional matrix, or tree). In aspects, the entries of network information can include respective PLMN codes of the networks (e.g., broken out into MCCs **302** and MNCs **304**) or respective network types **306** of the networks. The device settings **308** can indicate one or more wireless services that the wireless device is enabled to request or disabled from requesting. In aspects, the device settings **308** can include a visual, auditory, or haptic characteristic of the wireless device.

[0049] Although the device configuration data **300** includes specific examples of entries of network information and associated device settings, other examples are considered. For example, the device configuration data **300** can exclude the MCCs **302**, the MNCs **304**, or the network types **306**. In some cases, networks of the network types **306** can have the same settings. Thus, the different MCCs **302** and MNCs **304** can be redundant and can be absent from the device configuration data **300**. In this way, the device settings **308** can be associated with the network types **306**.

Alternatively, the device settings **308** can be associated with specific MCCs **302** and MNCs **304**. In

this way, different networks of the same network type can be associated with different device settings.

[0050] In some embodiments, the device configuration data **300** can be associated with a particular network provider. For example, the entries in the device configuration data **300** can have MNCs **304** associated with T-Mobile. Thus, the device configuration data **300** can provide the device settings **308** for the various networks associated with the particular network provider. The device configuration data **300** can be provided by the particular network provider (e.g., through a SIM card, over a wireless network, or otherwise). Given that the device configuration data **300** includes only data relating to the particular network provider, in some cases, the device configuration data **300** is utilized to only determine the device settings **308** in response to determining that the network information is associated with the particular network provider.

[0051] The example device configuration data **300** illustrated in FIG. 3 includes four entries of network information. Specifically, a first entry of network information has an MCC of **262**, which is associated with the country Germany, and an MNC of **01**, which is associated with T-Mobile in Germany. The first entry of network information further includes a network type that identifies the network as a terrestrial long-term evolution (LTE) network. Thus, the network associated with the first entry of network information can be an LTE terrestrial network operated by T-Mobile and located in Germany. Given that LTE terrestrial networks are often not resource-constrained, the device settings associated with the first entry of network information can allow the wireless device to request message, voice call, and data services. Moreover, the LTE wireless device can display an LTE icon to indicate that the wireless device is connected with an LTE network.

[0052] Similar to the first entry of network information, the second entry of network information indicates that the associated network is a terrestrial network. However, the network associated with the second entry of network information can be indicated as a 5G network. Moreover, the MCC, **310**, of the second entry of network indication indicates that the associated network is located in the United States, and the MNC, **660**, indicates that the associated network is operated by T-Mobile in the United States. Given that the network is a terrestrial network that is likely not resource-constrained, the device settings associated with the second entry of network information can have message, voice call, and data services enabled. Further, the device settings can specify that the wireless device is to display an icon associated with a connection to a 5G network.

[0053] The third entry of network information is associated with a non-terrestrial network. In aspects, the non-terrestrial network can be a non-terrestrial network partnered with T-Mobile to provide wireless communication services. To enable the non-terrestrial network to be identified by its PLMN code, T-Mobile can allocate a PLMN code associated with T-Mobile to the non-terrestrial network. For example, T-Mobile can provide wireless devices (e.g., subscribed to T-Mobile) with the device configuration data **300** that indicates that a network having an MCC of **312** associated with the United States and an MNC of **190** associated with T-Mobile is a non-terrestrial network. The device configuration data **300** can further define the device settings with which a wireless device is to comply when connected with the network associated with the third entry of network information. As illustrated, the device settings enable the wireless device to request message services from the non-terrestrial network but can disable the wireless device from requesting voice call and data services from the non-terrestrial network due to the increased resources required for these services and the decreased resources available on the non-terrestrial network. Similarly, the device settings can cause the wireless device to display an icon on the display of the wireless device that indicates that the wireless device is connected with the non-terrestrial network.

[0054] The fourth entry of network information is similarly associated with a non-terrestrial network. In aspects, the non-terrestrial network associated with the fourth entry of network information can be partnered with a network provider (e.g., T-Mobile) to provide a wireless communication service. In contrast to the third entry of network information, the fourth entry of

network information can have a PLMN code associated with a generic international network instead of a particular country. Moreover, the PLMN code can be associated with the particular network provider that provides the non-terrestrial network instead of a network provider with which the non-terrestrial network is partnered. As illustrated, the MCC of the non-terrestrial network is **901**, which is associated with international networks, and the MNC is **35**, which is associated with the network provider “Satellite Wireless.” To conserve resources on the non-terrestrial network, the device settings associated with the fourth entry of network information enable the wireless device to request message and voice call services but not data services. Moreover, the device settings can cause the wireless device to indicate that it is connected to the non-terrestrial network. As illustrated, the device settings associated with the fourth entry of network information cause the wireless device to output a sound that indicates that the wireless device is connected to the non-terrestrial network.

[0055] As illustrated, in some cases, the non-terrestrial networks can be indicated within the device configuration data **300** as having a PLMN code that belongs to and is allocated by a partnered network provider. In other cases, the non-terrestrial networks can have their own separate PLMN code. Moreover, the device configuration data **300** can specify device settings for each network type or for each network. Thus, in some embodiments, the individual network-specific information (e.g., MCC or MNC) or the network type can be omitted from the device configuration data.

[0056] In general, the device configuration data **300** can include other network information capable of being used to identify a network with which a wireless device can connect. Moreover, the device configuration data **300** can include other device settings. For example, the device configuration data **300** can include device settings that cause the wireless device to alter one or more characteristics of the wireless device, enable/disable one or more features on the wireless device, adjust one or more parameters defining communication between the wireless device and a mobile network, and so on.

Smart Device Application Prioritization on Satellite Network

[0057] FIG. **4** is a block diagram that illustrates an example of the disclosed technology **400** wherein the wireless device **404** restricts access to and prioritizes applications when connected to the non-terrestrial communication network **406**. The wireless device **404** disconnects from a terrestrial communication network **402** and connects to a non-terrestrial communication network **406**. The non-terrestrial communication network **406** provides a wireless communication service to the wireless device **404**. For example, the non-terrestrial communication network **406** can be a satellite that supplies wireless communication services.

[0058] In one example, the non-terrestrial communication network **406** transmits a list of resource constraints **408** to the wireless device **404**. The resource constraints **408** indicate to the wireless device **404** the resources and services available on the non-terrestrial communication network **406**. For example, the resource constraints **408** can indicate to the wireless device **404** the threshold under which applications must fall to comply with the resource constraints **408**. In another example, the resource constraints **408** can indicate to the wireless device **404** the amount of resources available on the non-terrestrial communication network **406**. In one example, the amount of resource constraints **408** can be dependent on the location of the wireless device **404** and/or the number of wireless devices connected to the non-terrestrial communication network **406**. In another example, the resource constraints **408** can indicate to the wireless device **404** to prioritize more resource-intensive communication applications while restricting lower resource-intensive applications such as those used for entertainment. The wireless device **404** can modify device settings to comply with the resource constraints **408**.

[0059] The wireless device **404** determines a list of applications found on the wireless device that comply with the received resource constraints **408**. The wireless device **404** restricts the ability to access applications that exceed the resource constraints **408** and becomes the restricted wireless device **410**. The restricted wireless device **410** does not have the ability to access applications that

do not comply with the resource constraints **408**. In one example, the restricted wireless device **412** can update the list of accessible applications based on changes to the resource constraints **408** of the non-terrestrial communication network **406**.

[0060] The restricted wireless device prioritizes the remaining applications that comply with the resource constraints **408** and becomes the prioritized wireless device **412**. The prioritized wireless device **412** contains only applications that comply with the resource constraints **408**. For example, the prioritization of the applications can be based on the type of application, the amount of resources the application requires, the recency the application was accessed, and/or the frequency the application is accessed. In one example, the prioritized wireless device **412** prioritizes contextual applications that provide information about the surrounding geographical area. In another example, the prioritized wireless device **412** prioritizes communication-based applications used for communicating with devices connected to a communication network. In another example, the prioritized wireless device **412** can update the prioritization based on changes to the resource constraints **408** of the non-terrestrial communication network **406**.

[0061] FIG. **5** is a flowchart that illustrates process **500** for dynamically restricting and prioritizing applications on a wireless device when the wireless device connects to a non-terrestrial communication network. In one example, the wireless device includes at least one hardware processor and at least one non-transitory memory storing instructions, which, when executed by the at least one hardware processor, cause the system to perform the process **500**.

[0062] At **502**, the wireless device can connect to a non-terrestrial communication network from a terrestrial communication network such that a wireless communication service is provided by the non-terrestrial communication network.

[0063] At **504**, the wireless device can receive an indication of resource constraints of the non-terrestrial communication network from the non-terrestrial communication network. In one example, the amount of resource constraints on the non-terrestrial communication network is based on a location of the wireless device and/or an amount of wireless devices connected to the non-terrestrial communication network. In another example, when the wireless device is connected to the terrestrial communication network, the wireless device can determine that the wireless device is traveling to a location serviced only by a non-terrestrial communication network. The determination can be based on current and previous location data. The wireless device can receive the indication of the resource constraints of the non-terrestrial communication network from the terrestrial communication network. The wireless device can download a contextual application not contained on the wireless device from the terrestrial communication network, where the contextual application complies with the resource constraints of the non-terrestrial communication network.

[0064] At **506**, the wireless device can determine a list of applications on the wireless device that comply with the resource constraints of the non-terrestrial communication network.

[0065] At **508**, the wireless device can restrict access to the applications that exceed the resource constraints of the non-terrestrial communication network. In one example, the wireless device can determine an alternative application for the wireless device to access. The alternative application complies with the resource constraints of the non-terrestrial communication network, and the alternative application is a less resource-intensive version of an application to be accessed by the wireless device. The wireless device can prioritize the alternative application.

[0066] At **510**, the wireless device can prioritize the applications on the wireless device that comply with the resource constraints of the non-terrestrial communication network. In one example, the prioritization is unique to each wireless device, location, or time. In another example, the wireless device prioritizes contextual applications based on a type of activity performed at the location of the wireless device. In another example, the applications are prioritized based on recency and frequency of being accessed on the wireless device. In another example, the wireless device prioritizes communication-based applications used for communicating with other wireless devices or emergency personnel.

[0067] At **512**, the wireless device can update the list of applications and prioritization of each application based on received changes to the resource constraints of the non-terrestrial communication network. In one example, the wireless device can generate a notification based on an update to the resource constraints of the non-terrestrial communication network, where the notification contains an updated list of applications the wireless device can access. The wireless device can notify the wireless device of the update to the list of applications that comply with the resource constraints of the non-terrestrial communication network.

Computing System

[0068] FIG. **6** is a block diagram that illustrates an example of a computer system **600** in which at least some operations described herein can be implemented. As shown, the computer system **600** can include: one or more processors **602**, main memory **606**, non-volatile memory **610**, a network interface device **612**, a video display device **618**, an input/output device **620**, a control device **622** (e.g., keyboard and pointing device), a drive unit **624** that includes a machine-readable (storage) medium **626**, and a signal generation device **630** that are communicatively connected to a bus **616**. The bus **616** represents one or more physical buses and/or point-to-point connections that are connected by appropriate bridges, adapters, or controllers. Various common components (e.g., cache memory) are omitted from FIG. **6** for brevity. Instead, the computer system **600** is intended to illustrate a hardware device on which components illustrated or described relative to the examples of the figures and any other components described in this specification can be implemented.

[0069] The computer system **600** can take any suitable physical form. For example, the computing system **600** can share a similar architecture as that of a server computer, personal computer (PC), tablet computer, mobile telephone, game console, music player, wearable electronic device, network-connected (“smart”) device (e.g., a television or home assistant device), AR/VR systems (e.g., head-mounted display), or any electronic device capable of executing a set of instructions that specify action(s) to be taken by the computing system **600**. In some implementations, the computer system **600** can be an embedded computer system, a system-on-chip (SOC), a single-board computer system (SBC), or a distributed system such as a mesh of computer systems, or it can include one or more cloud components in one or more networks. Where appropriate, one or more computer systems **600** can perform operations in real time, in near real time, or in batch mode.

[0070] The network interface device **612** enables the computing system **600** to mediate data in a network **614** with an entity that is external to the computing system **600** through any communication protocol supported by the computing system **600** and the external entity. Examples of the network interface device **612** include a network adapter card, a wireless network interface card, a router, an access point, a wireless router, a switch, a multilayer switch, a protocol converter, a gateway, a bridge, a bridge router, a hub, a digital media receiver, and/or a repeater, as well as all wireless elements noted herein.

[0071] The memory (e.g., main memory **606**, non-volatile memory **610**, machine-readable medium **626**) can be local, remote, or distributed. Although shown as a single medium, the machine-readable medium **626** can include multiple media (e.g., a centralized/distributed database and/or associated caches and servers) that store one or more sets of instructions **628**. The machine-readable medium **626** can include any medium that is capable of storing, encoding, or carrying a set of instructions for execution by the computing system **600**. The machine-readable medium **626** can be non-transitory or comprise a non-transitory device. In this context, a non-transitory storage medium can include a device that is tangible, meaning that the device has a concrete physical form, although the device can change its physical state. Thus, for example, non-transitory refers to a device remaining tangible despite this change in state.

[0072] Although implementations have been described in the context of fully functioning computing devices, the various examples are capable of being distributed as a program product in a variety of forms. Examples of machine-readable storage media, machine-readable media, or

computer-readable media include recordable-type media such as volatile and non-volatile memory **610**, removable flash memory, hard disk drives, optical disks, and transmission-type media such as digital and analog communication links.

[0073] In general, the routines executed to implement examples herein can be implemented as part of an operating system or a specific application, component, program, object, module, or sequence of instructions (collectively referred to as “computer programs”). The computer programs typically comprise one or more instructions (e.g., instructions **604**, **608**, **628**) set at various times in various memory and storage devices in computing device(s). When read and executed by the processor **602**, the instruction(s) cause the computing system **600** to perform operations to execute elements involving the various aspects of the disclosure.

Remarks

[0074] The terms “example,” “embodiment,” and “implementation” are used interchangeably. For example, references to “one example” or “an example” in the disclosure can be, but not necessarily are, references to the same implementation; and such references mean at least one of the implementations. The appearances of the phrase “in one example” are not necessarily all referring to the same example, nor are separate or alternative examples mutually exclusive of other examples. A feature, structure, or characteristic described in connection with an example can be included in another example of the disclosure. Moreover, various features are described that can be exhibited by some examples and not by others. Similarly, various requirements are described that can be requirements for some examples but not for other examples.

[0075] The terminology used herein should be interpreted in its broadest reasonable manner, even though it is being used in conjunction with certain specific examples of the invention. The terms used in the disclosure generally have their ordinary meanings in the relevant technical art, within the context of the disclosure, and in the specific context where each term is used. A recital of alternative language or synonyms does not exclude the use of other synonyms. Special significance should not be placed upon whether or not a term is elaborated or discussed herein. The use of highlighting has no influence on the scope and meaning of a term. Further, it will be appreciated that the same thing can be said in more than one way.

[0076] Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense—that is to say, in the sense of “including, but not limited to.” As used herein, the terms “connected,” “coupled,” and any variants thereof mean any connection or coupling, either direct or indirect, between two or more elements; the coupling or connection between the elements can be physical, logical, or a combination thereof. Additionally, the words “herein,” “above,” “below,” and words of similar import can refer to this application as a whole and not to any particular portions of this application. Where context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number, respectively. The word “or” in reference to a list of two or more items covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list. The term “module” refers broadly to software components, firmware components, and/or hardware components.

[0077] While specific examples of technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the invention, as those skilled in the relevant art will recognize. For example, while processes or blocks are presented in a given order, alternative implementations can perform routines having steps, or employ systems having blocks, in a different order, and some processes or blocks may be deleted, moved, added, subdivided, combined, and/or modified to provide alternative or sub-combinations. Each of these processes or blocks can be implemented in a variety of different ways. Also, while processes or blocks are at times shown as being performed in series, these processes or blocks can instead be performed or implemented in parallel or can be performed at different times. Further, any specific numbers noted

herein are only examples such that alternative implementations can employ differing values or ranges.

[0078] Details of the disclosed implementations can vary considerably in specific implementations while still being encompassed by the disclosed teachings. As noted above, particular terminology used when describing features or aspects of the invention should not be taken to imply that the terminology is being redefined herein to be restricted to any specific characteristics, features, or aspects of the invention with which that terminology is associated. In general, the terms used in the following claims should not be construed to limit the invention to the specific examples disclosed herein, unless the above Detailed Description explicitly defines such terms. Accordingly, the actual scope of the invention encompasses not only the disclosed examples but also all equivalent ways of practicing or implementing the invention under the claims. Some alternative implementations can include additional elements to those implementations described above or include fewer elements.

[0079] Any patents and applications and other references noted above, and any that may be listed in accompanying filing papers, are incorporated herein by reference in their entireties, except for any subject matter disclaimers or disavowals, and except to the extent that the incorporated material is inconsistent with the express disclosure herein, in which case the language in this disclosure controls. Aspects of the invention can be modified to employ the systems, functions, and concepts of the various references described above to provide yet further implementations of the invention.

[0080] To reduce the number of claims, certain implementations are presented below in certain claim forms, but the applicant contemplates various aspects of an invention in other forms. For example, aspects of a claim can be recited in a means-plus-function form or in other forms, such as being embodied in a computer-readable medium. A claim intended to be interpreted as a means-plus-function claim will use the words “means for.” However, the use of the term “for” in any other context is not intended to invoke a similar interpretation. The applicant reserves the right to pursue such additional claim forms either in this application or in a continuing application.

Claims

1. A wireless device comprising: at least one hardware processor; and at least one non-transitory memory storing instructions that, when executed by the at least one hardware processor, cause the wireless device to: connect to a non-terrestrial communication network from a terrestrial communication network such that a wireless communication service is provided by the non-terrestrial communication network; receive an indication of resource constraints of the non-terrestrial communication network from the non-terrestrial communication network; determine a list of applications on the wireless device that comply with the resource constraints of the non-terrestrial communication network; restrict access to the applications that exceed the resource constraints of the non-terrestrial communication network; prioritize the applications on the wireless device that comply with the resource constraints of the non-terrestrial communication network, wherein the prioritization is unique to each wireless device, location of the wireless device, or a time when the wireless device is accessed; and update the list of applications and prioritization of each application based on received indications of changes to the resource constraints of the non-terrestrial communication network.

2. The wireless device of claim 1, wherein the resource constraints on the non-terrestrial communication network is based on: the location of the wireless device; and an amount of wireless devices connected to the non-terrestrial communication network.

3. The wireless device of claim 1, wherein the instructions further cause the wireless device to: determine an alternative application for the wireless device to access, wherein the alternative application complies with the resource constraints of the non-terrestrial communication network, and wherein the alternative application is a less resource-intensive version of an application to be accessed by the wireless device; and prioritize the alternative application.

4. The wireless device of claim 1, wherein the wireless device prioritizes contextual applications based on a type of activity performed at the location of the wireless device.
5. The wireless device of claim 1, wherein the applications are prioritized based on recency and frequency of being accessed on the wireless device.
6. The wireless device of claim 1, wherein when the wireless device is connected to the terrestrial communication network, the instructions further cause the wireless device to: determine that the wireless device is traveling to a location serviced by the non-terrestrial communication network, wherein the determination is based on current and previous location data of the wireless device; receive the indication of the resource constraints of the non-terrestrial communication network from the terrestrial communication network; and download a contextual application not contained on the wireless device from the terrestrial communication network, wherein the contextual application complies with the resource constraints of the non-terrestrial communication network.
7. The wireless device of claim 1, wherein the instructions further cause the wireless device to: generate a notification based on an update to the resource constraints of the non-terrestrial communication network; wherein the notification contains an updated list of applications accessible by the wireless device; and notify the wireless device of the updated list of applications that comply with the resource constraints of the non-terrestrial communication network.
8. The wireless device of claim 1, wherein the wireless device prioritizes communication-based applications used for communicating with other wireless devices or emergency personnel.
9. A non-transitory, computer-readable storage medium comprising instructions recorded thereon, where the instructions, when executed by at least one data processor of a wireless device, cause the wireless device to: connect to a non-terrestrial communication network from a terrestrial communication network such that a wireless communication service is provided by the non-terrestrial communication network; receive an indication of resource constraints of the non-terrestrial communication network from the non-terrestrial communication network; determine a list of applications on the wireless device that comply with the resource constraints of the non-terrestrial communication network; restrict access to the applications that exceed the resource constraints of the non-terrestrial communication network; prioritize the applications on the wireless device that comply with the resource constraints of the non-terrestrial communication network; and update the list of applications and prioritization of each application based on received changes to the resource constraints of the non-terrestrial communication network.
10. The wireless device of claim 9, wherein the resource constraints on the non-terrestrial communication network is based on: a location of the wireless device; and an amount of wireless devices connected to the non-terrestrial communication network.
11. The wireless device of claim 9, wherein the instructions further cause the wireless device to: determine an alternative application for the wireless device to access, wherein the alternative application complies with the resource constraints of the non-terrestrial communication network, and wherein the alternative application is a less resource-intensive version of an application to be accessed by the wireless device; and prioritize the alternative application.
12. The wireless device of claim 9, wherein the applications are prioritized based on recency and frequency of being accessed on the wireless device.
13. The wireless device of claim 9, wherein when the wireless device is connected to the terrestrial communication network, the instructions further cause the wireless device to: determine that the wireless device is traveling to a location serviced by the non-terrestrial communication network, wherein the determination is based on current and previous location data of the wireless device; receive the indication of the resource constraints of the non-terrestrial communication network from the terrestrial communication network; and download a contextual application not contained on the wireless device from the terrestrial communication network, wherein the contextual application complies with the resource constraints of the non-terrestrial communication network.
14. The wireless device of claim 9, wherein the instructions further cause the wireless device to:

generate a notification based on an update to the resource constraints of the non-terrestrial communication network; wherein the notification contains an updated list of applications accessible by the wireless device; and notify the wireless device of the updated list of applications that comply with the resource constraints of the non-terrestrial communication network.

15. A method comprising: connecting to a non-terrestrial communication network from a terrestrial communication network such that a wireless communication service is provided by the non-terrestrial communication network; receiving an indication of resource constraints of the non-terrestrial communication network from the non-terrestrial communication network; determining a list of applications on a wireless device that comply with the resource constraints of the non-terrestrial communication network; restricting access to the applications that exceed the resource constraints of the non-terrestrial communication network; prioritizing the applications on the wireless device that comply with the resource constraints of the non-terrestrial communication network; and updating the list of applications and prioritization of each application based on received changes to the resource constraints of the non-terrestrial communication network.

16. The method of claim 15, further comprising: determining an alternative application for the wireless device to access, wherein the alternative application complies with the resource constraints of the non-terrestrial communication network, and wherein the alternative application is a less resource-intensive version of an application to be accessed by the wireless device; and prioritizing the alternative application.

17. The method of claim 15, wherein the wireless device prioritizes contextual applications based on a type of activity performed at a location of the wireless device.

18. The method of claim 15, wherein the applications are prioritized based on recency and frequency of being accessed on the wireless device.

19. The method of claim 15, wherein when the wireless device is connected to the terrestrial communication network, further comprising: determining that the wireless device is traveling to a location serviced by the non-terrestrial communication network, wherein the determination is based on current and previous location data of the wireless device; receiving the indication of the resource constraints of the non-terrestrial communication network from the terrestrial communication network; and downloading a contextual application not contained on the wireless device from the terrestrial communication network, wherein the contextual application complies with the resource constraints of the non-terrestrial communication network.

20. The method of claim 15, wherein the wireless device prioritizes communication-based applications used for communicating with other wireless devices or emergency personnel.
